

2 Starch molecules are the main storage molecules in many types of cereal grain, such as the grain of the barley plant.

(a) When the seed inside a barley grain germinates, genes coding for digestive enzymes are switched on. The enzymes that are synthesised catalyse the hydrolysis of storage molecules such as proteins and starch.

(i) Explain what is meant by a gene.

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.....
..... [2]

(ii) The hydrolysis of proteins in the barley seed produces amino acids that can be used in the synthesis of the proteins required for formation of the seedling (young plant).

Fig. 2.1 is an incomplete diagram of the molecular structure of the smallest amino acid, glycine. Each molecule of glycine has two carbon atoms.

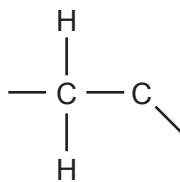


Fig. 2.1

Complete Fig. 2.1 to show the molecular structure of glycine. [2]

(iii) Starch is a mixture of two different molecules.

Name these **two** molecules.

.....
..... [1]

Two of the enzymes synthesised by the barley seed are α -amylase and maltase. These are involved in the hydrolysis of the stored starch during seedling formation.

In the food industry, the starch extracted from barley seeds (barley starch) is used in the production of sugar syrups. Fig. 2.2 summarises the reactions catalysed by α -amylase in the production of maltose syrup and by maltase in the production of glucose syrup.

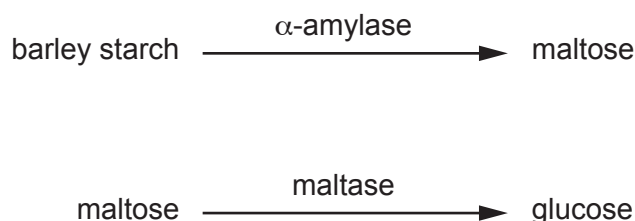


Fig. 2.2

(b) Some of the substances shown in Fig. 2.2 are listed in Table 2.1.

Complete Table 2.1 to identify which of the terms polysaccharide, monosaccharide and macromolecule apply to each of the substances listed.

Use a tick (✓) if the term applies and a cross (x) if the term does not apply.

Put a tick (✓) or a cross (x) in every box.

Table 2.1

substance	polysaccharide	monosaccharide	macromolecule
glucose			
maltase			
maltose			
starch			

[3]

When producing sugar syrups, there are advantages in using enzymes extracted from microorganisms.

For example, some enzymes extracted from microorganisms are heat stable. Heat-stable enzymes are used to increase productivity because the reactions can be carried out at higher temperatures.

(c) Suggest **one other** advantage of using enzymes obtained from microorganisms, rather than enzymes extracted from barley seeds, in the production of sugar syrups.

.....
 [1]

- (d) Fig. 2.3 is a graph showing how the activity of α -amylase extracted from barley seeds changes as the temperature increases from 10 °C to 66 °C.

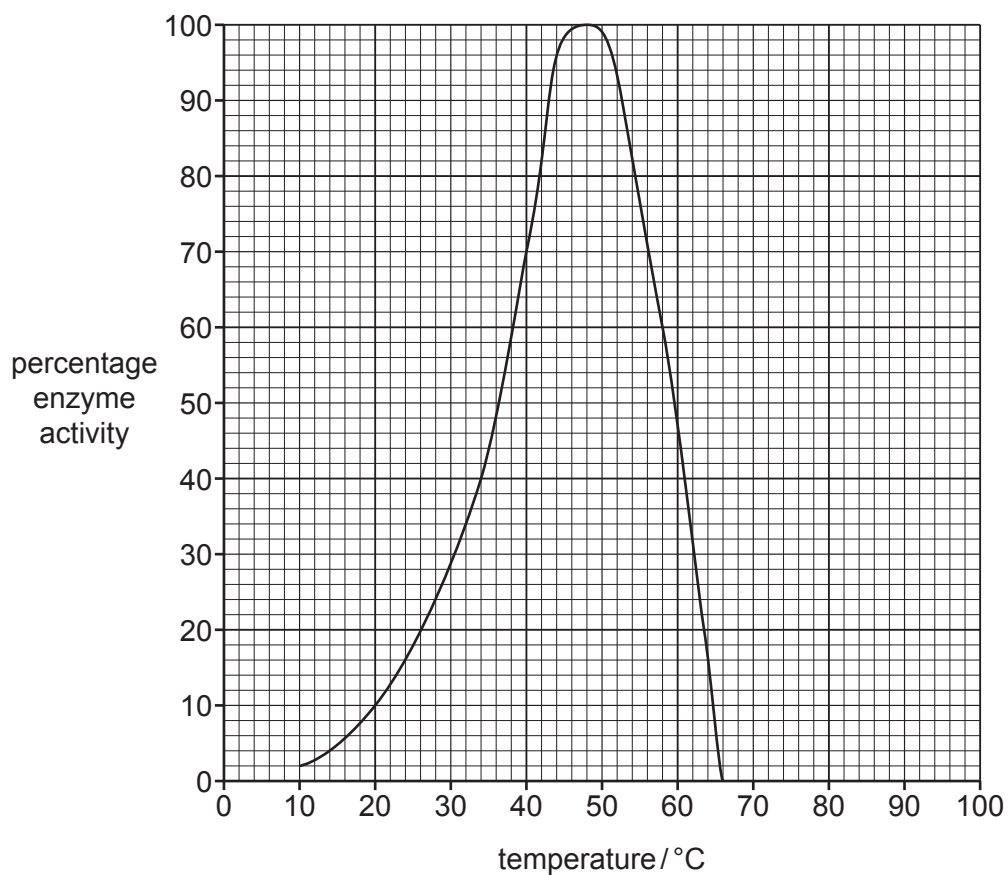


Fig. 2.3

- (i) Explain the effect of temperature on the activity of α -amylase extracted from barley seeds, as shown in Fig. 2.3.

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..... [3]

- (ii) Sketch on Fig. 2.3 the curve that would be obtained using α -amylase enzyme that is heat stable. [2]

[Total: 14]